



Influence of initial capital on optimal sizing of grid-connected photovoltaic system: A case study in Tunisia

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ABSTRACT

This research explores the incorporation of initial capital limitations into the design and optimization of grid-connected photovoltaic (PV) systems, with a focus on their economic and environmental implications. Moving beyond traditional approaches that prioritize energy cost reduction alone, this study presents a unique framework that harmonizes financial viability with technical efficiency. The investigation is divided into three main stages: determining system size, managing excess energy, and analyzing economic outcomes. A case study conducted in Sousse, Tunisia, validates the proposed methodology, utilizing simulation tools such as HOMER and MATLAB. The results demonstrate that an optimal PV setup, featuring a 2.25 kW photovoltaic array, a 1.5 kW inverter, and a 1 kWh battery, becomes economically viable when investments surpass TND 8 million. The findings reveal that integrating financial limitations enables a more strategic approach to renewable energy deployment, lowering the energy cost to TND 0.130/kWh and reducing CO₂ emissions from 2597 kg/year to 1490 kg/year. Additionally, the study offers insights applicable to other areas with comparable economic and environmental characteristics. This research provides a practical framework for policymakers and investors seeking to optimize PV systems under budget constraints, supporting the global shift toward sustainable energy practices.

Nomenclature

MG	Micro-Grid
PVP	Photovoltaic Panel
WT	Wind Turbine
DG	Diesel Generator
NPC	Net Present Cost
MPC	Model Predictive Control
TNPC	Total Net Present Cost
EC	Energy Cost
GA	Genetic Algorithm
HOGA	Hybrid Optimization Genetic Algorithm
STEG	Tunisian Company of Electricity and Gas
N_p	Number of Cells in Parallel
N_s	Number of Cells in Series

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T	Ambient Temperature
I_{pv}	PVP Current
I_{ph}	Photo-Current
I_{sat}	Saturation Current
V_{pv}	PVP voltage
P_{pv}	PVP output power
$P_{pv-rated}$	Rated power under reference conditions
I_s	Solar radiation
I_{s-ref}	Solar radiation under reference conditions
T_{ref}	Temperature under reference conditions
SOC	States Of Charge of battery
SOC_{max}	Maximal State Of Charge of battery
SOC_{min}	Minimal State Of Charge of battery
$P_{RS}(t)$	Power generated by renewable sources

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(continued)

$E_{RS}(t)$	Energy generated by renewable sources and stored in battery
$E_{pv}(t)$	Energy generated by PVP and stored in battery
E_{BT}	Energy stored in battery
E_{BTMin}	Minimal energy stored in battery
E_{BTMax}	Maximal energy stored in battery
P_{charg}	Power charging
E_{charg}	Energy charging
$P_{discharg}$	Power discharging
$E_{discharg}$	Energy discharging
$CO\&M$	Operation and maintenance cost
C_R	Replacement costs

1. Introduction

Currently, there is an urgent need to integrate the maximum amount of renewable energy resources to reduce harmful greenhouse gas emissions, voltage quality issues, power supply reliability, inflated network expansions, underserved remote areas and power losses, which exist in conventional power plants [1]. In Tunisia, almost all of the electrical energy is produced from crude oil, except for notable countries where electrical energy is produced from renewable sources. The cost of renewable energy is higher than that of traditional diesel generators. Nevertheless, Tunisia is characterized by a large amount of solar radiation (between 1800 (KWh)/m²/year in the north to 2600 kWh/m²/year in the south) and the price of solar panels has dropped worldwide [2].

Actually, investing in renewable energy sources, such as solar energy, can help reduce dependence on fossil fuels such as oil, hence increasing the country imports. Additionally, this can contribute to reducing CO₂ emissions, which can have economic and environmental benefits. The use of renewable energy can also lead to direct environmental benefits by reducing CO₂ emissions and limiting air, water and soil pollution [3–6]. Moreover, a combination of renewable energy sources can offer increased reliability of electricity supply, which is important for both grid-connected and off-grid systems. Investing in renewable energy sources can contribute to improving a country's economic and environmental sustainability. However, this often requires significant upfront investments, a long-term commitment to transitioning to cleaner and more sustainable energy sources.

Indeed, many countries have made considerable efforts to integrate grid-connected PV systems into traditional power plants. This integration can offer environmental benefits for the distribution society and customers. Customers who generate PV energy can sell their excess electricity to the grid, which can allow them to save money on their electricity bills and contribute to the production of green electricity [7]. Studies [8–10] have shown that annual PV electricity production on the grid is increasing and that this market is growing. This growth is due to the decrease in PV system costs and the solar energy support policies implemented by many countries. The integration of PV solar energy into power grids can also help reduce greenhouse gas emissions and improve energy security by diversifying sources of electricity supply. However, it should be noted that integrating grid-connected PV systems can also pose technical and regulatory challenges, such as managing energy production fluctuations and establishing a favorable regulatory framework.

Subsequently, the inverter size, storage capacity and optimization of PV arrays are central aspects of the design of grid-connected PV systems. The authors of [11] reviewed a variety of sizing techniques for PV systems, including grid-connected and standalone systems. The authors summarized that the availability of required technical data and traditional approaches (analytical, empirical, and numerical) provided valid results. Additionally, it was recommended that the nominal values of inverters installed with a grid-connected PV system would be lower than the size of the PVPs [8]. However, the optimal nominal value of the

converter may be the same or lower than the declared size of the PVPs [12]. This is fundamentally due to economic reasons, weather considerations, and the intrinsic parameters of the converter.

Although previous research ([8–10]) optimizes photovoltaic systems for cost or emissions, they frequently regard starting capital as a static quantity rather than a variable restriction. This paper presents a framework in which financial constraints directly affect component sizing and energy-selling techniques, highlighting novel trade-offs between investment levels and system viability. Our findings indicate that self-consumption (0 % grid sales) is best under capital limits, a conclusion not present in literature that presumes unlimited budgets.

Several grid-connected PV systems are not equipped with storage batteries. In such scenarios, excess electricity generated by the PV system is invested by selling it to the power grid at a predetermined price. Conversely, when the energy from the PVPs insufficient to power the primary load, the grid or storage devices (if available) can compensate for this energy deficiency [13].

Real-time pricing of electric energy during the integration of PV systems into the grid is based on adequate resource allocation. PVs are integrated into the grid on the basis of a properly distributed control framework. When the integrated system is operated to achieve optimal long-term and short-term management, the supervisory predictive model is used as a controller to manage energy intermittency, which is not within the scope of our current work.

Many studies on hybrid systems have been conducted for electrifying remote areas. In the literature, numerous researchers have investigated autonomous hybrid renewable energy sources to provide power to remote and rural areas. In Refs. [2,14], a PV-Wind Turbine-Battery Biomass system was analyzed in a Canadian urban area. In Tanzania, a standalone PV-Diesel Generator-Battery system was implemented [3–5]. A PV-WT system was proposed in Algeria [4,15]. In India and on Sultry Island, rural area electrification and small industry demand based on a PV-WT-Battery configuration were applied [7,8]. In Nigeria, six small urban residential areas were powered by a PV-DG-WT-Battery system [16]. The PV-DG-Biomass system was used to meet the energy needs of remote areas in India [17]. In the Philippines, a PV-WT-Battery-Biomass system was introduced with production and emission for a remote site [18]. To generate electricity, hydrogen and thermal supplies, the combination of a PV-DG-WT-Battery-Electrolyze-Boiler was used. Additionally, an economic representation of a Model Predictive Control (MPC) strategy with an external sizing loop based on a Genetic Algorithm (GA) and an internal MPC loop based on mixed-integer linear programming was discussed [14–16]. The advanced control method known as Model Predictive Control (MPC) makes use of a mathematical model of a system to forecast its future behavior and optimize its control actions over a limited time horizon.

In [12], the authors concluded that for a PV-WT-Battery hybridization, a 2 KW wind turbine mix was superior to standalone wind units and was more efficient than the solar system.

Meanwhile, another study demonstrated that the capacity of the wind turbine did not affect the simulation results [10]. In Egypt, for desalination and pumping applications, a PV-Battery-Fuel Cell hybrid plant was compared to generators and grid extensions. It was concluded that the projected system was more cost-effective than the grid extension [13].

In Iran, a biomass-wind power plant was designed with a capacity of 1.5 MW. The proposed system could be operated either as a standalone or grid-connected system [15]. In Ref. [19], it was shown that the share of renewable energy would increase to 68.5 % when a PV-Diesel-Battery-Wind system was integrated.

In a rural area of India, a PV-Battery-WT-DG system was planned to be used with a logistic growth model to predict future demand. The results indicated that the annual fixed cost of increasing production was achievable without significant environmental effects [17]. Similarly, the PV-DG-Battery-WT system in Malaysia was implemented with a higher energy cost (\$1.88/kWh) for small public customers with a peak load of

3.90 kW [15,20–22]. PV-Fuel Cell-WT residential systems were studied in Turkey for seasonal homes, where it was demonstrated that electrical energy storage systems were even better than hydrogen storage technologies [15].

A Proportional Integral (PI) control strategy taking into account the state of load of the battery was examined [20]. Similarly, in Iraq, a feasibility study of a grid-connected PV system was conducted. The analysis was carried out using HOMER software for four different regions to meet demand even in the event of a network energy shortage. It was demonstrated that the insertion of a DG would increase CO₂ emissions but reduce the NPC by 12 % [21]. The NPC is a financial statistic that determines the entire cost of an investment or project in terms of present value. Adjusted for the time value of money, it comprises all expenses incurred during the course of the project, including capital, operating, and future costs. Similarly, nine case studies were considered to electrify off-grid regions in Pakistan with an optimal sized Hybrid Renewable Energy System (HRES) containing PV/WT/DG and batteries.

At the same time, the intermittent nature of the energy produced by the HRES and the system harmonics were reduced in order to reduce costs. MPC based on MATLAB was used as well. This configuration enabled the reduction of pollutants, minimized the Cost of Energy (COE) and the Net Present Cost (NPC) [15,22,23].

The techno-economic features of PVP/WT/DG/battery and PVP/WT/battery hybrid power systems were assessed by Baseer et al. [24] in three residential neighborhoods of the Saudi Arabian industrial metropolis of Al Jubail in HOMER. The PVP/WT/DG/battery system was found to have the lowest COE. Three residential complexes with one, two, and three bedrooms that had daily load demands of 3288, 4865, and 11,160 kWh and peak loads of 270, 463, and 685 kW, respectively, were designed to be powered by this hybrid system.

The techno-economic viability of a hybrid WT/battery/DG/PVP power system was examined by Mamaghani et al. [25] in HOMER in three isolated Colombian settlements with varying climates. The best hybrid system in Puerto Estrella was determined to have a COE of 0.437 USD/kWh and an NPC of 836,210 USD.

The performance of a hybrid WT/DG/PVP power system for a Manitoba, Canada, home was evaluated by Arabzadeh Saheli et al. [26] using HOMER. With a total NPC of 33,110 USD and a COE of 0.444 USD/kWh, the ideal system had a 2-kW DG, five batteries, a 1-kW PVP system, and a 2-kW converter. According to the numerical data, the system reduced greenhouse gas emissions and was the most cost-effective and efficient of the systems studied.

For substantial non-residential power demand in the southern region of Iran, Baneshi and Hadianfard [27] used HOMER to perform a techno-economic analysis of off- and on-grid hybrid WT/PVP/DG/battery power systems. Off-grid hybrid systems' COE and renewable portion (RF) were determined to be 9.3–12.6 USD/kWh and 0 %–4.39 %, respectively. The grid under examination had an RF of 0 %–53 % and a COE of 5.7–8.4 USD/kWh.

The performance of an off-grid hybrid WT/DG/battery/PVP system in a remote location was examined by Shezan et al. [28] using HOMER. When compared to conventional power systems, the optimized hybrid system was shown to have a about 29.65 % lower NPC and 16 tons reduced CO₂ emissions. The optimized system has a COE of 1.877 USD/kWh and an NPC of about 288.194 USD.

In contrast to Refs. [24,28], which optimize systems for pre-determined budgets, our research demonstrates that financial limitations reshape ideal designs—specifically, self-consumption (0 % grid sales) is feasible only when initial capital surpasses TND 8 million (Section 5).

The incorporation of renewable energy, especially photovoltaic (PV) systems, into national energy grids is a vital strategy for lowering carbon emissions and advancing energy sustainability. Solar energy has become a preferred choice for many nations, driven by its declining costs and positive environmental impact. Nevertheless, even as PV technology becomes more affordable, the upfront investment needed for installation

continues to pose a major challenge, particularly in developing economies. These financial limitations can affect the design and economic viability of PV projects, adding complexity to investment planning and decision-making.

While significant research has been dedicated to optimizing grid-connected PV systems, the majority of studies concentrate on reducing energy costs, enhancing system efficiency, or lowering CO₂ emissions. However, the role of initial capital investment in shaping system design and ensuring long-term financial sustainability has received limited attention. In contrast to prior research, which often emphasizes system sizing based on energy balance and cost optimization alone, this study integrates the effects of initial financial limitations into the optimization framework. Furthermore, by conducting a comparative analysis of various energy selling scenarios, the research offers a more comprehensive understanding of the economic viability and environmental benefits of grid-connected PV systems.

Tunisia presents an excellent case study for this research, given its rich solar resources and active efforts toward energy transition. Although the country enjoys optimal solar conditions, the widespread adoption of PV systems is frequently constrained by financial challenges. By examining the balance between upfront investment, energy costs, and environmental advantages, this study offers critical insights that extend beyond Tunisia, providing a framework for other regions with comparable economic and climatic profiles.

Current research on grid-connected photovoltaic systems mostly emphasizes techno-economic optimization (e.g., minimizing net present cost or cost of energy) while considering initial capital as a constant or subordinate factor. Baseer et al. [24] and Mamaghani et al. [25] optimize hybrid systems for cost-effectiveness without regard to budget constraints, whereas Shezan et al. [28] emphasize emissions reduction under unrestricted financial conditions. Likewise, Chen et al. [13] and Luoma et al. [12] examine inverter sizing and grid interaction while presuming unlimited investment capabilities. This oversight is crucial for practical implementations, because initial capital limitations immediately affect system scalability and profitability. Our research tackles this deficiency by presenting a dynamic financial constraint paradigm, wherein optimal component sizing (PV, battery, inverter) and energy-selling tactics are directly linked to investment levels. This technique, as illustrated in Sections 4–5, uncovers non-linear trade-offs, including the TND 8 million criterion for viability, that have not been examined in previous work.

Although numerous studies have optimized photovoltaic system sizing based on technical or economic criteria like cost of energy (COE) and net present cost (NPC), few have explicitly incorporated starting capital as a dynamic optimization constraint. Baseer et al. [24] and Chen et al. [13] presume the availability of complete funds and do not evaluate the viability of scenarios constrained by investment limitations. Likewise, techniques reliant on post-optimization financial screening fail to adjust component sizing or energy strategies according to capital limits.

Our research fills this gap by considering the available budget as a fundamental design motivator, enabling us to reveal how particular investment levels transform optimal topologies, such as the transition from grid sale to self-consumption.

Although several studies have concentrated on reducing the cost of energy (COE) or optimizing the net present cost (NPC), few have explicitly incorporated the initial capital requirement as a critical restriction in the sizing process. Classical methodologies typically presume an unrestricted investment capacity or focus exclusively on technical or economic parameters for optimization. Nonetheless, in underdeveloped nations or for individuals, the scarcity of accessible capital is a significant and constraining barrier. This study is characterised by the explicit incorporation of this limitation during the optimization phase, enabling the suggestion of realistic, feasible, and adaptable designs based on varying degrees of initial investment.

This study seeks to optimize the design of a grid-connected PV system

by integrating initial capital constraints into the decision-making framework.

The primary contributions of this research can be summarized as follows. First, we propose a groundbreaking optimization framework that incorporates financial limitations into PV system design, enabling practical and economically feasible implementation. Second, we develop a systematic methodology that harmonizes system performance, investment costs, and environmental outcomes through integrated simulations using HOMER and MATLAB.

Third, we present a case study with wide-ranging relevance, providing actionable insights for policymakers and investors in areas facing comparable financial and energy-related obstacles. Finally, we conduct a detailed comparative assessment of system designs, analyzing how varying investment levels influence energy costs, system efficiency, and reductions in CO₂ emissions.

The structure of this paper is organized as follows: Section 2 outlines the mathematical modeling of the PV system, incorporating both economic and environmental factors. Section 3 introduces the optimization methodology, utilizing HOMER and MATLAB for analysis. Section 4 provides an overview of the input data and case study specifications. Section 5 presents the simulation outcomes and a comparative evaluation of the results. Finally, Section 6 summarizes the study's key findings and suggests potential directions for future research.

2. Mathematical modeling of the PV system

The developed mathematical model takes into account several key parameters to evaluate the performance of the PV system on the grid, including the current, the voltage, the battery State of Charge (SOC), the thermal energy consumption by network providers, the hourly exchanged energy, and the economic estimates. These parameters allow quantifying the system performance in different scenarios and determining the optimal sizes of system components to produce a minimum unit energy rate for a minimum budget cost.

Understanding the performance of the PV system on the electrical grid is crucial as it enables determining the system efficiency and measuring its impact on the electrical grid. Developing an equivalent electrical model allows simulating and predicting the behavior of the PV system on the electrical grid.

By focusing on the current, the voltage and the battery (SOC), it is possible to determine how the PV system interacts with the electrical grid and measure the system efficiency in terms of energy production.

The thermal energy consumption by grid suppliers is also important to consider as it can affect the energy production of the PV system due to ambient temperature. If the ambient temperature is high, the energy production of the PV system may be reduced due to the decreased efficiency of solar cells.

The hourly exchanged energy is an important indicator of the performance of the PV system on the electrical grid, as it enables measuring the system efficiency in terms of supplying energy to the grid and energy consumption by grid users.

Finally, the economic estimates are essential to measure the cost of the PV system and its impact on the electricity cost for the grid users. The costs may include installation costs, maintenance costs, operating costs, as well as the cost of the electricity generated by the PV system.

Overall, developing an equivalent electrical model and considering these various variables allow measuring the efficiency and economic impact of the PV system on the electrical grid, which is essential for planning and optimizing large-scale solar energy production systems.

It is important to note that for PV systems, no fuel is required for electricity production. The annual revenue for a PV project is typically determined by the amount of electricity produced and sold to the grid, as well as the savings realized on electricity costs through the use of solar energy. Annual expenses include the operating and maintenance costs of the system and the initial installation costs. Therefore, fuel depletion is not a relevant parameter for evaluating the economic profitability of a

PV project. The key parameters for assessing the economic profitability of a PV project are typically the levelized COE and NPC.

However, the actual cost of fuel supplied to thermal power plants is estimated based on the Franco on board oil price, the transportation cost which also depends on the distance between the fuel source and the plant, the cost of investment in tank trucks, and the availability of trucks. Expenses related to thermal power plants are ignored when the system is autonomous in order to estimate the savings of the grid-connected PV system. On the other hand, the overall bill is included to fully evaluate the grid-connected project by considering fuel as one of the input data.

The developed configuration is designed to optimize the sizing of the components based on the NPC, the COE and the CO₂ of the PV system when an electrical load is connected to the STEG thermal power stations under different weather conditions.

The proposed mathematical model uniquely integrates initial capital constraints into the optimization framework. By combining HOMER for energy system simulations and MATLAB for cost and emissions analysis, this research enables a thorough assessment of system performance across different financial scenarios. This dual-tool approach ensures a holistic evaluation, balancing technical, economic, and environmental factors.

2.1. Modeling electric parameters

The maximum energy generated by the PVs (PG-max) is expressed as follows [29,30]:

$$P_{G-\max} = P_c \frac{GSR}{FR} (1 + P_t (T_p - T_{pref})) P_{G-sdc} \quad (1)$$

where GSR represents the global solar radiation, T_p is the temperature of the PV arrays, FR is the irradiance of the flux, P is a factor that depends on Joule effects, cleanliness and instability of the PV panels, P_t is the temperature coefficient, T_{pref} and P_{G-sdc} are respectively the ambient temperature (25 °C) and the nominal power of the PV generator in a standard test environment (1000W/m²).

The ambient temperature T_p is influenced by the intensity of GSR and the ambient temperature T_a [25].

$$T_p = T_a + \frac{GSR}{800} (SCOT - 20) \quad (2)$$

SCOT represents the operating temperature under standard conditions. These conditions are the ambient temperature of 20 °C, the average wind speed of 1 m/s, and the irradiance of 800 W/m² [31,32].

2.2. Economic modeling

The profitability of the PV system that is connected to a thermal power station is evaluated by estimating the annual cost of the Cost, invest project, which is considered as the net present value. The total annual cost depends on the annual cost of Operation and Maintenance (O&M) and the annual cost of investment [33].

Therefore, the annual cost of the project is the difference between the first term, which represents the savings obtained by connecting the system to the grid, and the second term, which represents the costs spent on this project.

In our work, the electrical system that provides the smallest Net Present Cost (NPC) (as shown in Eq. (3)) is identified as the best configuration. The NPC determines the total costs and benefits over the lifetime of the device.

$$NPC = \frac{C_{tot,ann}}{CRF_{(i,T_p)}} \quad (3)$$

Where $C_{tot,ann}$ represents the annual cost (TND/year), i represents the real interest rate (%) and T_p illustrates the project lifetime (in years).

The capital recovery factor (CRF) is a ratio employed to calculate the present value of a series of uniform yearly cash flows. Mathematically, the CRF is represented as follows:

$$CRF(i, n) = \frac{i(i+1)^n}{(i+1)^n - 1} \quad (4)$$

n represents the number of years.

The Recovery Cost (RC) is the derived value of the remaining components of the system at the end of the project life. The SC (Salvage Cost) is calculated as shown in Eq. (5).

$$SC = C_{RC} \frac{T_{rem}}{T_{com}} \quad (5)$$

Where T_{com} , C_{RC} and T_{rem} are respectively the replacement cost (TND), the remaining life of the components (in years) and the life of the components (in years).

The cost of energy COE is given by Eq. (6). It represents the average cost per kWh of actual electricity produced by the energy system.

$$COE = \frac{C_{tot,ann}}{E_{tot,ann}} \quad (6)$$

Where $E_{tot,ann}$, is the total annual electricity produced by serving the load and the amount of electricity sold to the grid (kWh/year).

3. Optimization methodology using HOMER and MATLAB

Unlike traditional methods that presume boundless capital and assess financial feasibility post-system size, our methodology incorporates initial capital availability immediately into the sizing process. This facilitates a dynamic modification of photovoltaic, inverter, and battery capacities in accordance with budget limitations, yielding innovative insights into the relationship between economics and system design.

Optimizing a grid-connected photovoltaic (PV) system demands a holistic approach that combines economic, technical, and environmental considerations. This research utilizes HOMER (Hybrid Optimization of Multiple Energy Resources) and MATLAB to identify the ideal system setup that reduces the cost of energy (COE), net present cost (NPC), and CO₂ emissions while maintaining energy reliability.

In contrast to conventional HOMER simulations that presume static budgets, our MATLAB-enhanced system iteratively modifies component dimensions (PV, battery, inverter) to conform to capital constraints. This demonstrates non-linear correlations between investment and COE/NPC, as illustrated in Figs. 8–10, where a TND 8 million threshold transitions the ideal system from grid sales to self-consumption.

The optimization process follows a series of structured steps to determine the most effective PV system configuration.

3.1. Input data collection

To ensure a realistic and context-specific system design, several essential input parameters were considered in this study.

- ✓ **Load Profile:** Energy demand data is gathered for the chosen location, accounting for both seasonal and daily fluctuations.
- ✓ **Meteorological Data:** Solar radiation and temperature data are sourced from NASA databases and local weather stations.
- ✓ **Economic Parameters:** Costs associated with PV panels, inverters, batteries, maintenance, and grid electricity rates are factored into the analysis.
- ✓ **System Constraints:** Key limitations, such as battery storage capacity, grid reliability, and inverter efficiency, are clearly defined.

3.2. Simulation and optimization in HOMER

HOMER evaluates multiple system configurations by adjusting PV capacity, battery storage, and inverter size. Sensitivity study assesses both technical parameters and investment limits (e.g., TND 2M to TND 10M), demonstrating how capital constraints limit component sizing—a factor overlooked in Refs. [13,24].

The optimization process relies on the following steps.

- ✓ **Economic Metrics:** The configuration with the lowest net present cost (NPC) and cost of energy (COE) is selected as the optimal solution.
- ✓ **Sensitivity Analysis:** The effects of changes in initial capital, energy prices, and component costs are thoroughly assessed.
- ✓ **Scenario Evaluation:** Various rates of surplus energy sales (ranging from 0 % to 100 %) are analyzed to identify the most economically viable strategy.

3.3. Advanced refinement in MATLAB

MATLAB enhances HOMER outputs by simulating capital-constrained dynamics: at each investment level (TND X), battery charging and discharging patterns, along with grid sales, are adjusted to maximize ROI under budgetary constraints, differing from the static analysis presented in Refs. [12,25].

After generating initial results from HOMER, MATLAB is employed to further enhance the analysis.

- ✓ **Curve Fitting and Trend Analysis:** MATLAB is used to establish relationships between initial capital investment and key performance metrics, such as the cost of energy (COE), net present cost (NPC), and CO₂ emissions.
- ✓ **Dynamic Energy Flow Modeling:** The tool optimizes the charging and discharging patterns of batteries, as well as their interaction with the grid, to enhance system efficiency.
- ✓ **Grid Dependency Assessment:** MATLAB evaluates the effects of minimizing reliance on grid electricity, providing insights into the system's autonomy and sustainability.

4. Input data and case study parameters

The load profile for this study is derived from actual electricity consumption data from a residential area in Sousse, Tunisia. The peak power demand reaches 3.17 kW, with an average daily energy consumption of 11.26 kWh. The electrical load profile was created using HOMER software, taking into account seasonal fluctuations. Table 1 details the appliances included, their power ratings, and operating hours. Fig. 1 visually represents the daily and seasonal variations in electricity demand, highlighting patterns and trends.

4.1. Load demand data

It is quite common for the electricity consumption of a region to vary

Table 1
Load request in one day in July.

Loads	Number in use	Power (w)	Operating hours	Load (w/hours)
Lighting	7	10	7	490
Television	1	300	6	1800
Radio	1	10	9	90
Air conditioner	3	3000	12	108000
Computer	1	45	10	450
Phone load	3	10	3	90
Blender	1	400	0,3	120
Refrigerator	1	600	24	14400

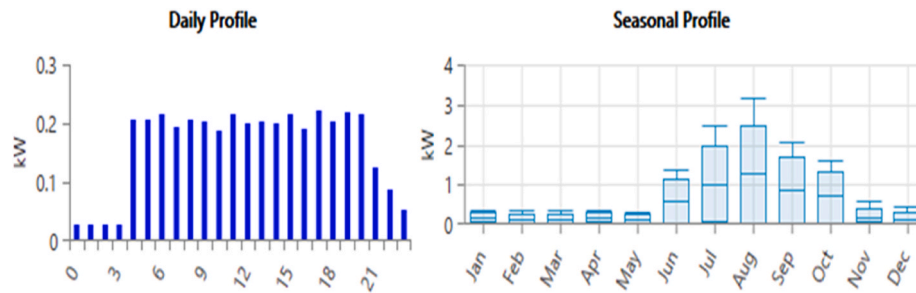


Fig. 1. A summary of the load's daily and seasonal load profiles, as well as its minimum and maximum values.

depending on various factors, such as climate change. Electricity demand data can be collected over different periods, such as monthly cycles, daily cycles or annual data, to better understand the region energy consumption. Fig. 1 shows the annual data.

For the Sousse region, the proposed load peak demand is 3.17 kW, which means that the system must be able to provide enough electricity to meet this maximal demand. Additionally, the average energy supply is 11.26 kWh per day, which must also be taken into account when planning the system.

Using the HOMER tool, it is possible to design a hybrid energy system tailored to the region energy consumption. Table 1 shows the primary base load created by HOMER, which is the minimal amount of energy required to power the region's basic needs in the absence of peak demand.

In summary, the use of tools such as HOMER allows for a better understanding of the energy needs of a region and designing hybrid energy systems tailored to those needs. This can help ensure a reliable and sustainable energy supply for local communities.

4.2. Solar irradiance

The meteorological data utilized in this study were gathered from a measurement station located at the Higher Institute of Applied Sciences

and Technology in Sousse, Tunisia (Fig. 2) [34,35]. The analysis incorporates real-time solar radiation and wind speed data to accurately assess the renewable energy potential. Solar radiation levels vary between 3.5 kWh/m²/day and 7 kWh/m²/day, with the highest irradiation observed from March to September (Fig. 3). Additionally, the wind speed profile, illustrated in Fig. 4, offers further insights into the region's renewable energy resources, complementing the solar data.

4.3. Energy rate

The economic constraints examined in this study encompass investment costs, replacement costs, and operational & maintenance (O&M) expenses for various system components. Table 2, Tables 3, and Table 4 provide a detailed breakdown of these costs for photovoltaic panels, converters, and battery storage systems, respectively. Furthermore, the cost of electricity purchased from the grid is established at 0.243 TND/kWh, while the selling price of surplus electricity is set at 0.073 TND/kWh (Table 5). These figures are derived from current market conditions and regulatory frameworks in Tunisia. The financial evaluation employs key indicators such as Net Present Cost (NPC) and Cost of Energy (COE) to assess the profitability of the system across different scenarios.

Excess electricity is supplied by the PV system to the public service

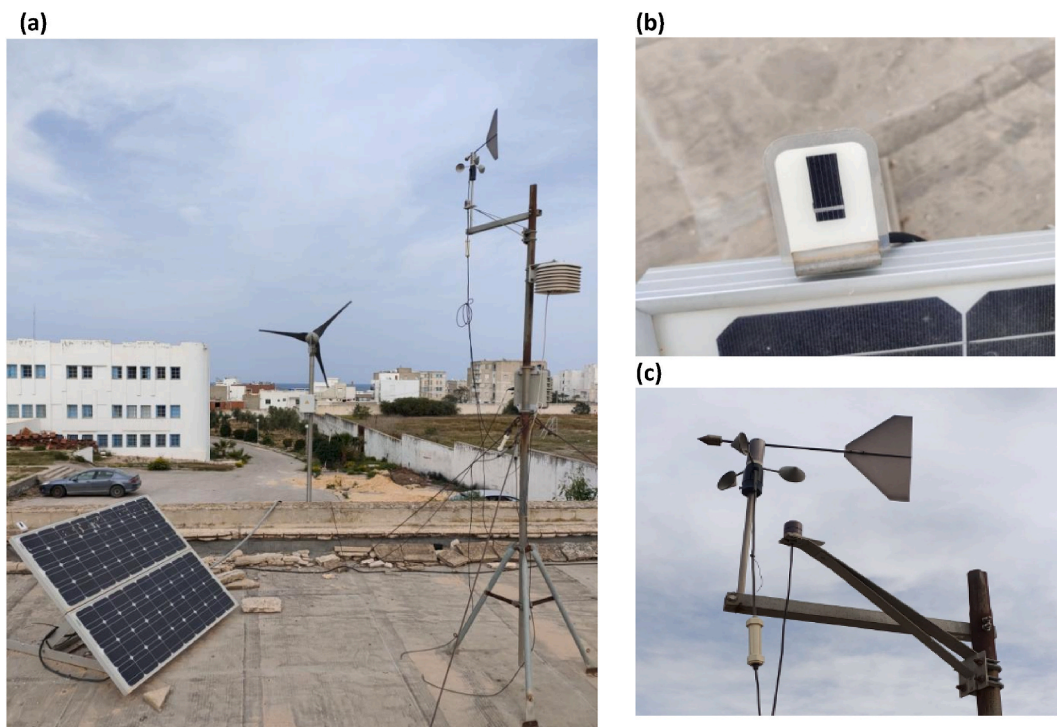


Fig. 2. (a): Hybrid PVP/WT system; (b): Solar radiation sensor; (c): Wind sensor.

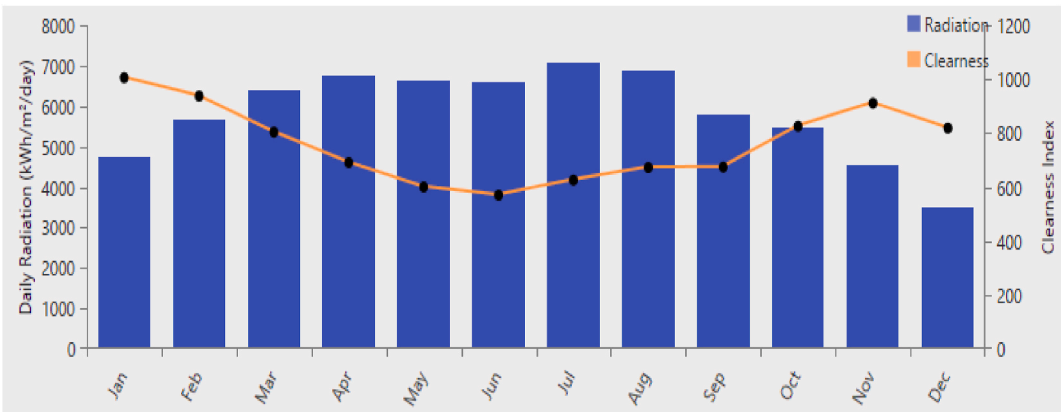


Fig. 3. Monthly profile of global solar radiation and brightness index for the city of Sousse-Tunisia.

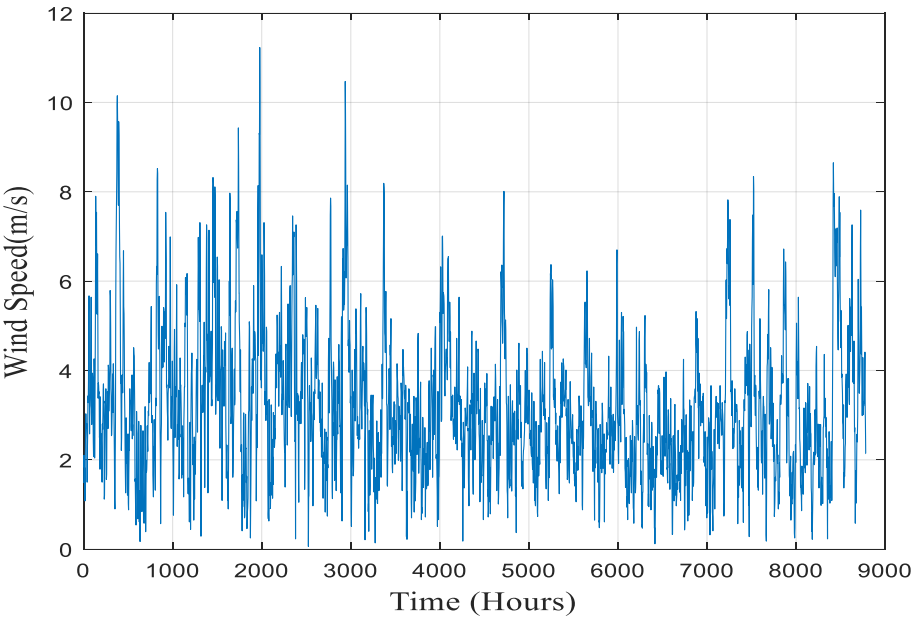


Fig. 4. Annual wind speed.

Table 2
Costs and technical parameters for PV module.

Description	Data
Replacement cost	920 TND/kW
Investment cost	1000 TND/kW
O&M cost	55 TND/year
Lifetime	25 years

Table 3
Converter data.

Description	Data
Replacement cost	75 TND/kW
Investment cost	100 TND/kW
O&M cost	16 TND/year
Efficiency	98 %
Lifetime	15 years

Table 4
Storage system data.

Description	Data
Declared voltage	12 VDC
load state	40 %
Replacement cost	300 TND/kW
Investment cost	350 TND/kW
O&M cost	10TNDyear

Table 5
Power grid purchased and sales prices (TND/KWh).

Specifications	Cost (TND)
Grid power cost (TND/KWh)	0.243
Grid sellback cost (TND/KWh)	0.073

when the purchase rate is higher than the selling rate [36]. In this study, excess electricity from the PV system is used to load the batteries when the batteries are not fully loaded and then sold to the public service to

reduce unused energy and achieve reasonable efficiency while reducing CO₂ emissions. Selling excess energy to the public service is part of the energy-saving regulations, and it is strongly recommended by Tunisia, which aims to reduce local fuel consumption. As a result, the environmental effects are minimized.

5. Simulation results and comparative analysis

The system considered in this study consists of a load, a converter and PVPs.

5.1. Design specifications

This system is connected to the public service according to the configuration depicted in Fig. 5. The objective of this system is to improve the cost of energy by using the electricity produced by the PVPs to power the load or sell it to the grid during periods of sunlight.

In addition to improving the cost of energy, this system also helps to reduce fluctuations in the output voltage of the PVPs, eliminate the phase difference between the photovoltaic panels and the grid voltage, and decrease issues related to the quality and reliability of energy. These advantages are important as they help to ensure a stable and reliable power supply for end-users and for the utility grid.

a. PV arrangements

It appears that the capacity of the PVPs in this study depends on factors such as solar irradiation, availability, demand characteristics, and the fraction of energy produced and sold to the public utility. The renewable energy produced by the PVPs is then used to meet the demand for electricity.

However, the production of the PVPs can be controlled based on the load requirements, in size of the PVPs, and the proportion of energy produced and sold. It seems that in this study, the electricity generated by the PV system is produced from 6 a.m. to 6 p.m.

The PV system data are presented in Table 2, where it appears that the operating cost is negligible and regular maintenance is performed at a very low cost. This can be beneficial for the overall profitability of the system and can help to reduce costs for end-users.

b Converter

It appears that in this project, the DC current produced by the PVPs is converted into AC current by an inverter, in order to power a given load. Converters are widely used because they are highly efficient and easy to install.

According to Table 3, the cost of the converter in this project is estimated at 75 TND/KW, based on the Tunisian market. The operating and maintenance cost of the converter is estimated at 16 TND/year. It is important to take these costs into account in the overall evaluation of the profitability of the PV system.

c Storage battery

The aim of using storage devices in hybrid systems is to inject the load with the necessary amount of energy when the PV output is zero. Additionally, it can support in the billing management. In this study, the battery system can be charged from the main grid or from PVs. Table 4 presents the data of the battery. Compared to the other system elements, the costs are high.

It appears that the purpose of using storage devices in hybrid systems is to provide the necessary amount of energy to the load when the output from the PV system is zero. This can help to ensure that the load is always supplied with electricity, even during periods of low solar irradiation. Additionally, the use of storage devices can assist in the billing management, by allowing excess energy to be stored for later use. Table 4 provides the data related to the battery system, and it suggested that the costs associated with this element of the system are relatively high compared to other system components. It is important to consider these costs when evaluating the overall economic viability of the hybrid system.

d Utility system

It appears that the electrical grid is used to meet the load requirements when the PV system cannot produce enough energy due to a lack of available solar irradiation. During the night, energy production by the PV system is zero, which means that the load must be supplied by the electrical grid.

In this study, it is indicated that the current technology leads to the use of a flexible tariff system, which is presented in Table 5. The electricity purchase rate from the grid is assumed to remain constant throughout the day, at an amount equal to 0.243 TND/kWh.

5.2. Simulation results and discussions

It appears that this study aims to optimally size a PV solar system with storage for an electrical grid in Tunisia, considering multiple sizes for the system components. However, the grid is modeled based on fuel finances, given that most stations in the Tunisian national grid are thermal power plants.

The algorithm illustrated in Fig. 6 represents the final phase of the optimal sizing analysis. In this phase, the best configuration is selected based on previously gathered data to maximize energy efficiency while minimizing the Cost of Energy (COE), Net Present Cost (NPC), and CO₂ emissions. For each time interval Δt , the algorithm uses inputs such as solar radiation, ambient temperature, power demand, energy sold, and

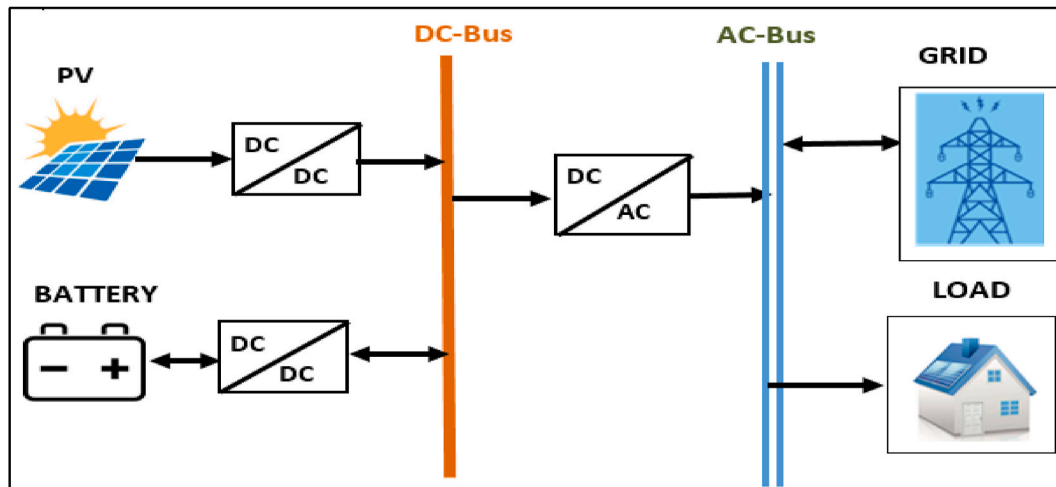


Fig. 5. Configuration of the grid-connected PV system with battery storage and inverter.

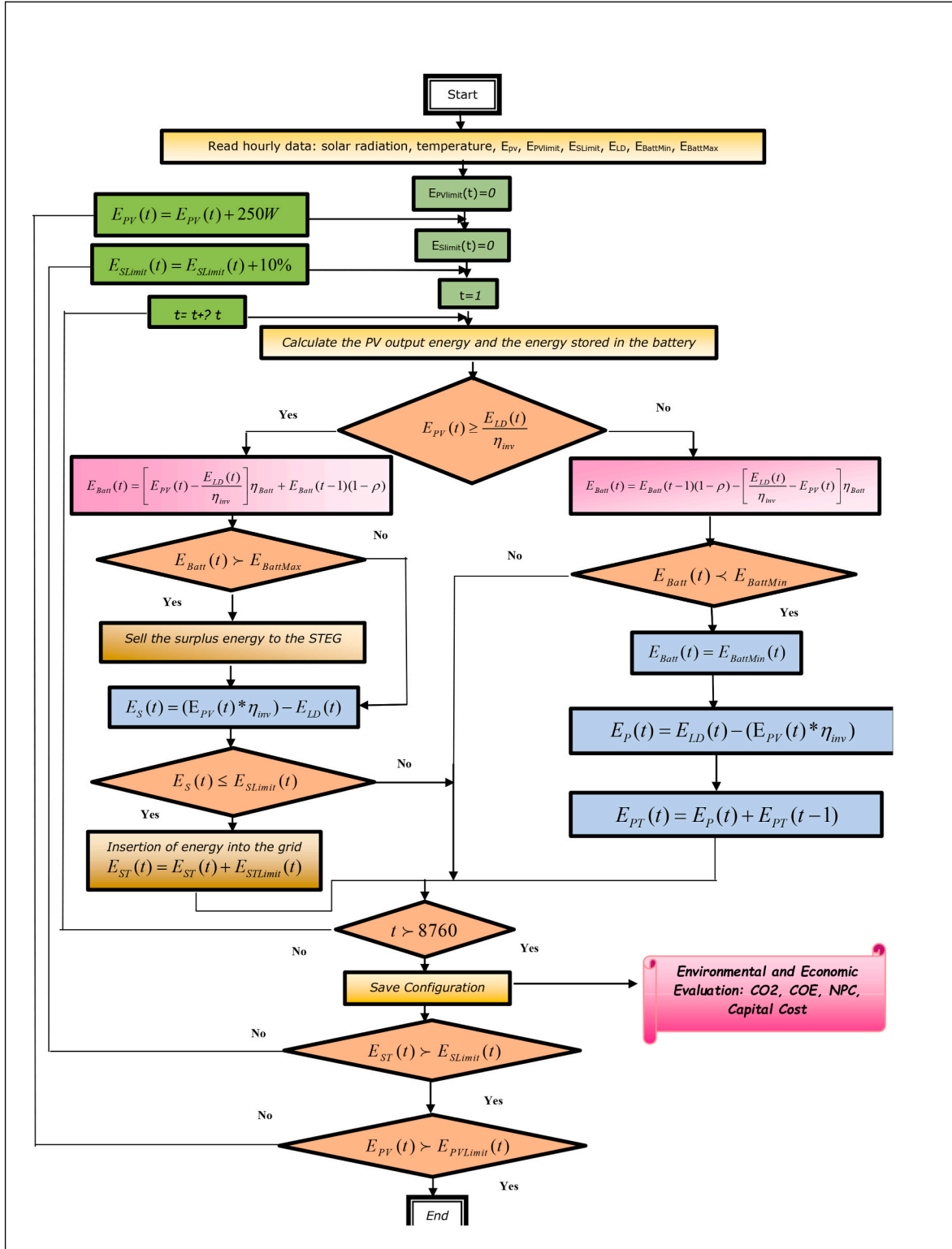


Fig. 6. Flowchart of the optimal sizing of the PVP/Inverter/Battery system based on the percentage of the quantity of electricity sold to STEG.

the maximum and minimum states of charge of the battery.

The methodology can be outlined in the following steps:

Case (a): If the energy produced by the photovoltaic system exceeds the load demand for a given hour, the surplus energy is stored in the battery bank. The updated state of charge is then calculated using the following equation:

$$E_{Batt}(t) = \left[E_{pv}(t) - \frac{E_{LD}(t)}{\eta_{inv}} \right] \eta_{Batt} + E_{Batt}(t-1)(1-\rho) \quad (7)$$

Case (b): When the hourly load demand exceeds the energy output from the photovoltaic system, the battery bank is used to meet the load requirements. The updated state of charge is then calculated using the following equation:

$$E_{Batt}(t) = E_{Batt}(t-1)(1-\rho) + \left[\frac{E_{LD}(t)}{\eta_{inv}} - E_{pv}(t) \right] \eta_{Batt} \quad (8)$$

In case (a), if the energy stored in the battery exceeds the maximum threshold, the surplus energy will be sold to STEG, and the energy sold $E_s(t)$ is given by the following equation:

$$E_s(t) = (E_{pv}(t) \times \eta_{inv}) - E_{LD}(t) \quad (9)$$

In case (b), if the energy stored in the battery is lower than the minimum threshold, the energy purchased $E_p(t)$ is given by this equation:

$$E_p(t) = E_{LD}(t) - (E_{pv}(t) \times \eta_{inv}) \quad (10)$$

A MATLAB program was developed to size the various components of the microgrid system based on the scenarios described above. This sizing process aims to meet the system's power requirements while minimizing CO₂ emissions over a specified period TTT (one year in this study) and achieving the lowest possible Net Present Cost (NPC) and Cost of Energy (COE).

Where: E_{LD} is the Energy load, E_p and E_{PT} represented respectively the Energy purchased and the Total Energy purchased.

E_s and E_{ST} represented respectively the Energy sold and the Total energy sold.

Further, the main objective of our optimal micro-grid system sizing algorithm, is to detect the optimal configuration providing the maximal energy with COE, NPC and CO₂ minimization. This algorithm uses the energy production of the renewable source and the battery.

The next steps are as follows.

- Give the energy production of the renewable source and the battery for one year.
- Check the charging or discharging mode according to the need for or surplus of energy.
- Using HOMER, determine the optimal system for each of the following cases: 0 %, 25 %, 50 %, 75 %, 100 % of energy sold to the public grid.
- Having determined the optimal system for each case, repeat the same process, but with the following condition: each system sells all its surplus energy to the STEG.
- Use MATLAB to plot the COE, NPC, CO₂ and NP curves as a function of the change in initial capital, in order to determine the best optimal system used by the STEG.

The system is also designed when excess no electricity is sold to the (STEG), and the system performance is evaluated in terms of the leveled cost of energy (COE), the emissions, the excess electricity, the annual electricity bill and the share of renewable energy.

It is important to note that the results of this study depend on the assumptions and parameters used, and they may vary based on the specific conditions of the electrical grid in Tunisia and the changes in equipment and energy costs. Therefore, it is essential to consider these factors when interpreting the results of the study.

Table 6 examines the effects of selling different percentages of photovoltaic electricity produced (0 %, 25 %, 50 %, 75 %, and 100 %). It

lists the various components of the optimal photovoltaic system for each energy sale rate, highlighting key specifications such as panel capacity, storage devices, and expected performance in terms of cost and energy efficiency.

Fig. 6 complements this analysis by illustrating the optimal sizing algorithm for each energy sale scenario described in Table 6. The algorithm outlines the successive steps required to determine the optimal configuration of the photovoltaic system, taking into account technical and economic parameters associated with the sale of surplus energy.

In this section, after determining the optimal system architecture for each sales percentage value (0 % up to 100 %), we move on to redo the simulation for each architecture but with all energy produced by the PVP is sold to the STEG. Table 7 shows that the values for COE, NPC and CO₂ emissions change significantly for all architectures when the rate of sale of surplus energy varies between 0 % and 75 %. However, these values remain stable when the rate reaches 100 %, indicating optimized performance in this specific case. In addition, this table confirms that the system integrating the total sale (100 %) of surplus energy is the one that offers the lowest overall cost, reinforcing its economic attractiveness.

These results are the result of optimal sizing, using the algorithm shown in Fig. 6. This algorithm, designed to adapt to different scenarios of energy sold, has been specifically modified for the case of a 100 % sale rate, by incorporating the instruction $EsLimit(t) = EsLimit(t) + 100$ %. This adaptation optimizes not only the technical configuration of the system, but also its economic performance, guaranteeing a sustainable and efficient solution for managing surplus energy.

The results presented in Table 7 show that the system with the largest installed capacity (PVP) is always the most profitable, with the lowest cost. This correlation highlights the economic advantage of high-capacity systems, which optimize available resources while minimizing the costs associated with energy production.

→ This interpretation, based on the results presented in Table 7, leads us to analyse the impact of increasing installed photovoltaic capacity. Such an approach would make it possible to integrate an increased number of subscribers (or projects), thereby increasing the initial capital invested. This study aims to assess the economic and technical implications of such an increase, while examining the potential benefits in terms of profitability and energy efficiency.

a. Simulation results

MATLAB software is used to present the variation in economic and ecological determinants of the system based on different sizes of the proposed system elements. However, the objective is to optimize the different sizes of system elements and reduce environmental impacts. The meteorological data used in this study is generated by specifying the latitude and longitude of the study site, while HOMER is able to link the location with NASA sources to report information on sunlight and temperature (excluded in this research) throughout the year [37].

However, the small size of the inverter increases the operating and maintenance costs of the grid-connected PV system, especially when its capacity is zero kW (i.e. when it is disconnected). Simulation shows that the current net cost is very high in this situation. The reason is that

Table 6
Configurations with different percentages of energy sold (0 %, 25 %, 50 %, 75 %, 100 %).

Grid sale capacity (%)	Grid sale capacity (kW)	Architecture				Cost				System	
		PV (kW)	Grid (kW)	Converter (kW)	Dispatch	NPC (TND)	COE (TND)	Operating cost (TND/yr)	Initial capital (TND)	Ren Frac (%)	CO ₂ (kg/yr)
0	0	2.25	999999	1.5	LF	10588.1	0.2015696	658.1291	2175	39.14518	1580.375
25	0.79	2.75	999999	1.5	LF	9208.207	0.1305686	514.9827	2625	57.32367	1487.975
50	1.585	3.5	999999	2	LF	8314.336	0.09612297	388.3437	3350	67.8505	1374.823
75	2.3775	3.75	999999	2.5	LF	7856.044	0.08199272	330.9807	3625	71.57433	1346.516
100	3.17	3.75	999999	2.5	LF	7855.414	0.0819794	330.9314	3625	71.57667	1346.516

Table 7

Configurations where all energy produced by the PVP is sold to STEG (100 %).

Grid sale capacity (kW)	Architecture				Cost				System	
	PV (kW)	Grid (kW)	Converter (kW)	Dispatch	NPC (TND)	COE (TND)	Operating cost (TND/yr)	Initial capital (TND)	Ren Frac (%)	CO ₂ (kg/yr)
total	2.25	999999	1.5	LF	8940.95	0.1222818	529.2781	2175	56.28143	1580.375
total	2.75	999999	1.5	LF	8698.819	0.1131315	475.135	2625	60.85768	1487.975
total	3	999999	2	LF	8308.489	0.09900554	423.0884	2900	65.38275	1436.241
total	3.5	999999	2	LF	8144.566	0.09190543	375.0632	3350	68.62037	1374.823
total	3.75	999999	2.5	LF	7855.414	0.0819794	330.9314	3625	71.57667	1346.516

electricity without the inverter is supplied directly by the thermal power grid, which results in a high bill. Additionally, the effect of the inverter size on CO₂ emissions is the same as on operating and maintenance costs. Similarly, the inverter shutdown significantly increases pollutants that are impacted by non-renewable energy sources. Moreover, excess electricity is also present when the inverter is used. Clearly, the converter is considered as a driving element for determining the economic and ecological aspects.

After having determined the values of each optimal system in the cases where the rate of sale is 100 %, the curves are plotted using MATLAB software.

b. Impact of change in initial capital on number of projects

Investing in small-scale projects offers many strategic advantages for STEG, including flexibility, risk spreading, gradual implementation and greater involvement of the local community. An analysis of Fig. 7 clearly shows that the number of projects in an optimal system with 0 % sales to the public grid increases significantly as initial capital grows, outperforming the other systems studied (25 %, 50 %, and 100 % sales to the grid).

This rapid increase in the number of projects in the 0 % sales scenario highlights the economic viability of this approach. It demonstrates that a model based solely on self-consumption is not only profitable but also better suited to meeting the growing demand for renewable energy, while requiring an initial investment that is less dependent on income from sales to the public grid.

What's more, this strategy offers additional advantages:

Risk spreading: By diversifying projects, the financial or technical impact of a potential failure is limited.

Gradual implementation: Allows continuous adaptation to changing needs and economic constraints, while offering opportunities for technological improvement.

Involvement of the local community: The realization of numerous

small-scale projects fosters the involvement and awareness of local communities in sustainable energy, while stimulating local economies.

In conclusion, the system based on a strategy of 0 % sales to the public grid, as shown in Fig. 7, is a promising approach that optimizes financial resources in an efficient and sustainable way, while meeting renewable energy development objectives.

c. Impact of size variation on COE

The results presented in Fig. 8 highlight the impact of the size of the photovoltaic system, inverter and storage, as well as the level of initial investment, on the average cost of energy (COE). A reduction in the size of these components, combined with an increase in the initial capital invested by STEG in the renewable energy installation project for homes connected to the public grid, leads to a significant reduction in the COE.

When applying the optimal system identified above, it appears that the most profitable scenario corresponds to a 0 % sales rate, i.e. a model of complete self-consumption with no energy sales to the public grid. This scenario is distinguished by a COE reaching a value of 0.130 TND, which represents a particularly competitive energy cost compared with the other systems evaluated.

Component reduction: Reducing the size of photovoltaic systems, inverters and storage devices contributes directly to lower installation and maintenance costs, positively influencing COE.

Increased initial investment: Increased initial capital enables STEG to achieve economies of scale, resulting in lower average long-term energy costs.

0 % grid sales scenario: This self-consumption model appears to be the most economically efficient, minimizing costs while maximizing benefits for end-users. The COE of 0.130 TND illustrates the effectiveness of this scenario in providing affordable renewable energy.

The 0 % sale to the public grid scenario not only optimizes energy costs, but is also part of a strategy to support the energy transition. By reducing dependence on the public grid and promoting self-

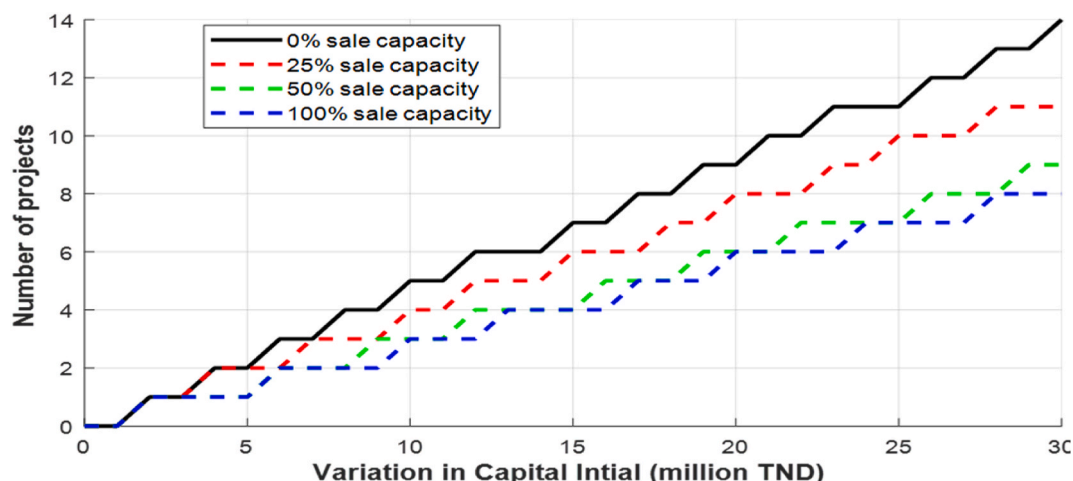


Fig. 7. Number of projects by diversity of initial capital (million TND).

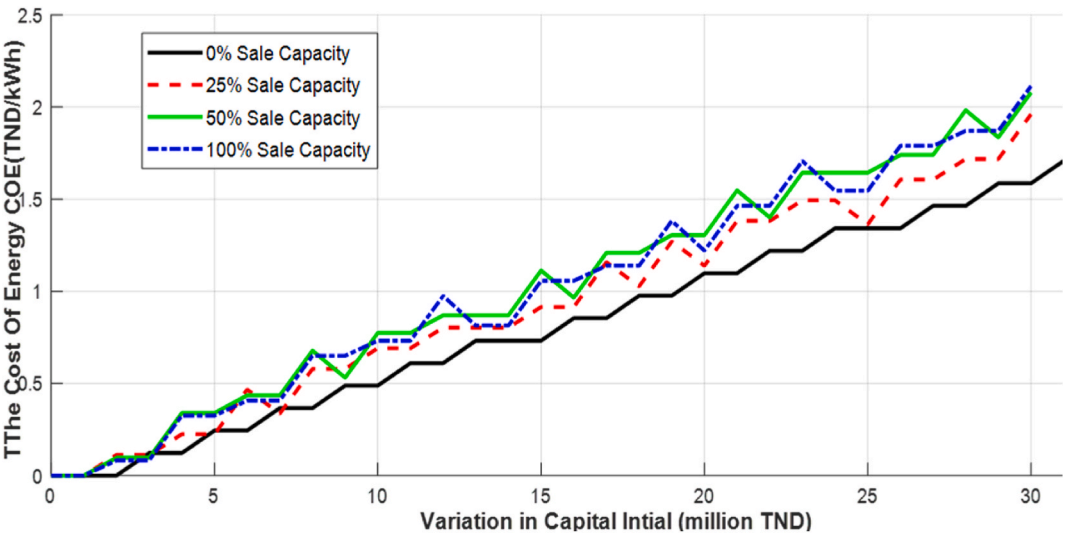


Fig. 8. COE (TND/kWh) according to different initial capital (million TND).

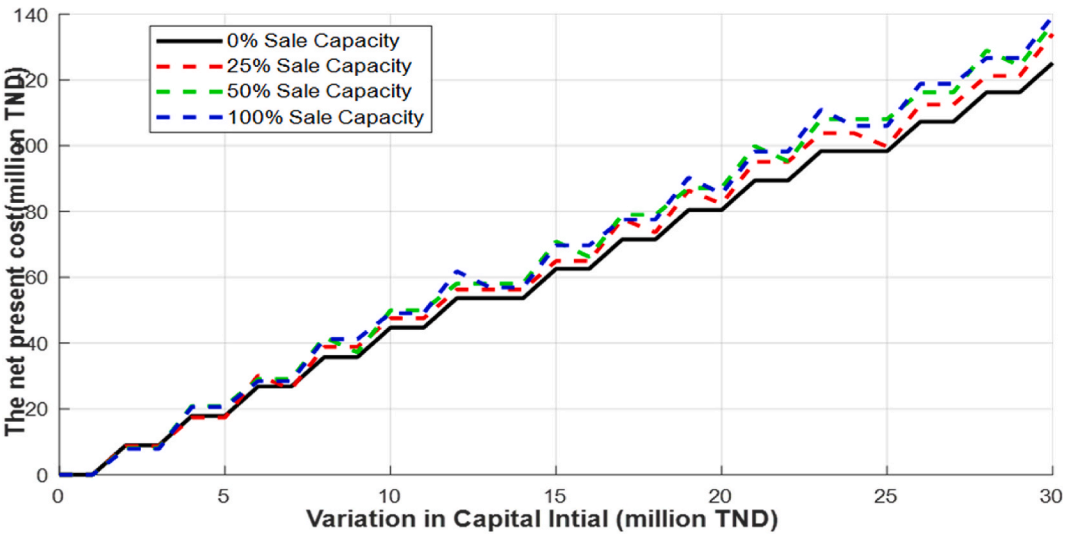


Fig. 9. NPC (million TND) by diversity of initial capital (million TND).

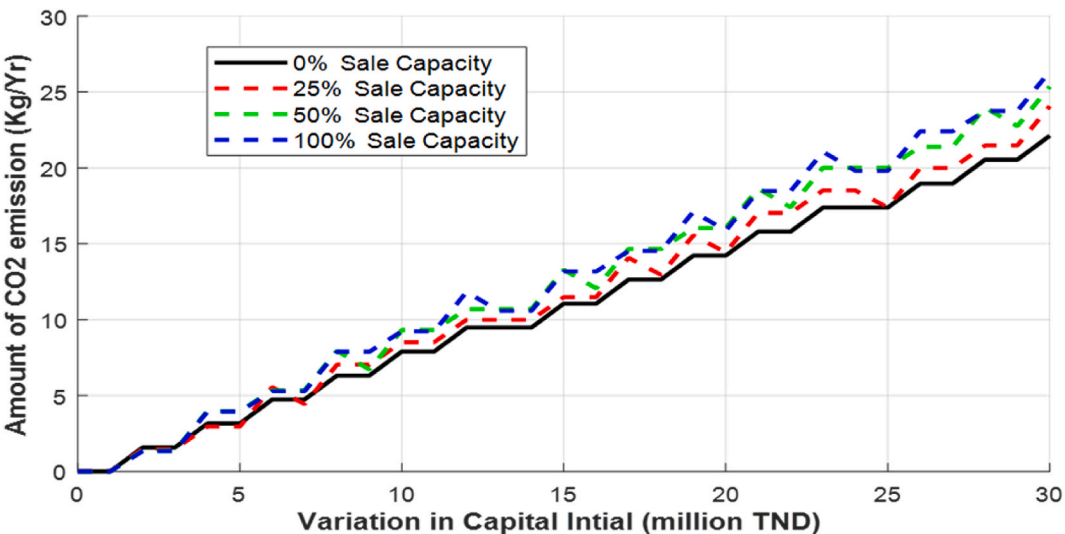


Fig. 10. CO₂ as a function of initial capital diversity (TND).

consumption, this approach contributes to a more sustainable and financially advantageous model for STEG and its customers. Fig. 8 validates the effectiveness of the optimal system in reducing COE.

d. Impact of size variation on NPC

The results presented in Fig. 9 highlight that the optimal system corresponding to 0 % sales to the public grid scenario, with a net present cost (NPC) of TND 7 million, is the most profitable of the configurations analyzed. This increased profitability is particularly noticeable as the initial capital invested increases, underlining the importance of substantial initial funding.

Analysis of results:

Increasing financial returns: The 0 % sales system demonstrates better financial performance than the other scenarios, as the higher initial investment enables overall costs to be optimized, notably through economies of scale.

Impact of initial investment: STEG's greater capital allocation to these projects encourages the installation of more efficient renewable energy infrastructure, thereby reducing long-term costs. This scenario highlights the importance of self-consumption in the business model, which eliminates the costs associated with selling energy to the public grid.

System viability and optimization: The optimal configuration with an NPC of TND 7 million validates the financial feasibility of the approach, while meeting sustainable development and renewable energy objectives.

These results clearly show that increasing the initial capital invested by STEG is a winning strategy for maximizing financial returns from renewable energy projects. The 0 % sales to the public grid scenario not only offer the best return on investment, but also contributes to a more sustainable energy transition independent of the public grid. Fig. 9 illustrates that adopting this model can be considered an economically and environmentally viable solution for grid-connected homes.

e The impact of the size variation on CO₂ emission

The optimization algorithm used in this analysis calculates the CO₂ emissions associated with each optimal system. This data is then used in MATLAB to identify the system best suited to STEG's needs. Fig. 10 illustrates a significant result: as initial investment increases in the 0 % grid sales scenario, the system achieves minimal levels of CO₂ emissions.

CO₂ emissions reduction: In the scenario without sales to the public grid, the amount of CO₂ emissions is estimated at 1490 kg/year, which is a minimum value.

This performance can be attributed to full self-consumption and the optimized use of renewable energies, reducing dependence on fossil fuels.

Impact of initial investment: An increase in initial investment encourages the adoption of more efficient and sustainable technologies, such as advanced photovoltaic panels and high-performance storage systems, contributing to a significant reduction in emissions.

Comparison with other scenarios: Scenarios involving sales to the public grid (25 %, 50 %, 100 %) generate slightly higher CO₂ emissions. This may be linked to a different system configuration or to inefficiencies due to the infrastructure required to sell energy.

Fig. 10 highlights that the 0 % sales to the public grid scenario, in addition to its financial advantage, is also the most favorable from an environmental point of view. With an annual emission of 1490 kg of CO₂, this system minimizes the ecological impact while meeting the energy needs of the homes. These results underline the importance for STEG of investing in self-consumption systems that combine economic profitability with environmental sustainability.

f Comparison investigation

The system proposed by HOMER and MATLAB is considered the best optimal system, which should be preferred over other combinations. When considering the investment capital parameter in the choice of the optimal system, as taken into account by the STEG, it is observed that raising the investment capital makes the optimal system with 0 % sales the most profitable solution. This conclusion is based on the results obtained from Figs. 7–9, and Fig. 10.

g Technical analysis of winning system

In the previous section, it has been shown that the system created using HOMER-MATLAB has the best economic performance. Therefore, this winning plan is studied in more detail with regard to its technical aspects.

By utilizing the HOMER simulation tool, an optimal system is proposed, which involves configuring various input parameters including the meteorological data from a house in Tunisia and the load profile. Four different combinations are presented in Table 8, and an economic evaluation is performed to identify the systems with the lowest energy cost. Subsequently, a detailed technical study is conducted on the selected systems.

To provide a comprehensive analysis of the system, five economic variables are considered: the NPC, the COE, the O&M costs, the replacement cost, and the recovery cost. This system configuration incorporates PVPs, a battery storage system and a converter, enabling seamless energy adaptation between the load demand, the PV battery and the grid supply. These components are interconnected to the public service, ensuring efficient energy utilization.

The suggested plan comprises PV-BSS battery-Converter elements, which are grid-connected. This combination ensures the provision of electricity to the targeted home load at the lowest energy tariff compared to other configurations, without considering efficiency. It offers the advantage of the lowest tariff in comparison to alternative configurations, while still delivering optimal power supply.

In Tunisia, imported electricity from the grid is cost-effective, and it has been proven that a storage system is necessary to regulate the load bus voltage. In our specific case, this configuration offers the most environmentally-friendly and economically viable operation. However, it should be noted that utilizing a battery system without PVPs results in the highest operating and maintenance costs.

As indicated in Table 8, removing the battery storage and relying solely on the PV system in the previous operational state leads to significantly higher operating and maintenance expenses. This, in turn, significantly reduces the levelized COE to its minimal value of 0.122 TND.

The table compares four energy system configurations in terms of net present cost (NPC), average cost of energy (COE), component capacity (PV, battery, converter) and share of renewables. System 1 (PV-Converter-Grid) is the most cost-effective, with an NPC of 8.941 MTND and a COE of 0.122 TND/kWh. This system uses photovoltaic panels connected to the public grid, offering a significant renewable energy share of 56.3 %. System 2 (PV-Battery-Converter-Grid), with the addition of a battery, slightly increases NPC (9.255 MTND) and COE (0.130 TND/kWh), while achieving the highest share of renewables (57.2 %). By contrast, Systems 3 and 4 (Battery-Grid and Grid) are far less competitive. System 3, without photovoltaic panels, has an NPC of 12.764 MTND and a COE of 0.243 TND/kWh, with a total absence of renewable energies. System 4, based entirely on the public grid, is the least efficient, with a high NPC of 13.661 MTND, a COE of 0.260 TND/kWh and a negligible contribution from renewables (0.0124 %). Overall, System 1 stands out as the most economical, while System 2 offers a good compromise between energy autonomy and profitability. Systems 3 and 4 are only suitable in specific cases where renewable energies are not an option.

The total electricity production is predominantly fueled by excess electricity and renewable energies, accounting for 85.7 % and 59.9 % of

Table 8

Presentation of different sales configurations at STEG.

System	Configuration	NPC (MTND)	COE (TND/kWh)	PV (kW)	Battery (kW)	Renewable (%)	Converter (kW)
1	PV-Converter -Grid	8941	0,122	2.25	–	56.3	1.5
2	PV-Battery-Converter-Grid	9255	0,130	2.25	1	57.2	1.5
3	Battery-Grid	12,764	0,243	–	–	0	
4	Grid	13,661	0,260	–	1	0.0124	1.5

the energy mix, as depicted in Fig. 11. This explains the significant amount of energy being directly sold to the grid, resulting in a noteworthy reduction in electricity costs. Moreover, nearly half of the electricity consumed is sourced from renewable origins, leading to positive environmental effects by mitigating harmful impacts and generating cost savings on electricity bills.

To elucidate the uniqueness of our capital-constrained optimization methodology, Table 9 delineates critical methodological and outcome distinctions between our study and previous studies that either presumed limitless budgets or regarded financial constraints as peripheral factors.

We can affirm that this study enhances previous research by illustrating how early capital limitations reshape optimal photovoltaic system design. In contrast to traditional methods (e.g., Refs. [13,22]), which see budgets as ancillary, our framework demonstrates that financial constraints require self-consumption models (0 % grid sales) to optimize returns within investment limits—an essential revelation for emerging countries.

Fig. 12 illustrates a concentration of energy consumption and demand rates during a specific period of the day, from 6:00 a.m. to 10:00 p.m. However, these rates vary depending on the season and they are measured in TND/kW-month. Specifically, during the summer season, electricity demand exceeds production rates, as highlighted in the load profile section.

This can be attributed to the need for powering various energy-intensive equipment and systems during these peak demand hours.

The utilization of renewable energy sources, such as PV solar energy and inverters, plays a vital role in reducing carbon dioxide (CO₂) emissions. Thanks to these technologies, annual CO₂ emissions have been reduced to 1.490 kg/Year, thus contributing to environmental preservation and the fight against climate changes.

The results demonstrate that higher initial capital investment leads to a substantial reduction in both the Net Present Cost (NPC) and Cost of Energy (COE), alongside a notable decrease in CO₂ emissions. The analysis reveals that a self-consumption model (0 % selling scenario) provides the optimal balance between economic viability and environmental advantages. This finding is particularly valuable for policy-makers seeking to encourage the adoption of renewable energy systems, as it highlights the importance of aligning financial strategies with sustainability goals.

In contrast to earlier studies that only emphasize lowering the LCOE or assessing various hybrid designs, our methodology incorporates a practical aspect: the immediate influence of initial capital limitations on

Table 9

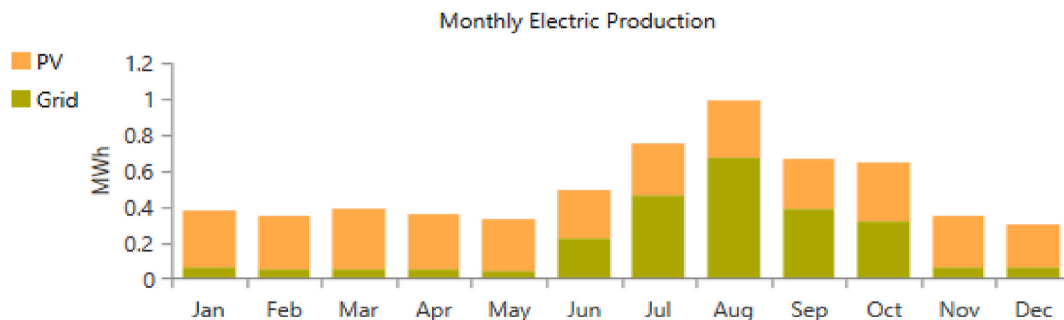
Comparative analysis of PV system optimization strategies—financial limitations against prior research.

Study	Optimization Focus	Financial Constraints?	Key Finding	Our Advancement
Baseer et al. [24]	Hybrid system COE	No	PV/WT/DG/battery = lowest COE	Budget not considered→ strategy shift
Chen et al. [13]	Inverter sizing	No	Sizing based on irradiation	Assumes full funding, no sizing trade-off
This study	Grid-connected PV	Yes (TND 8M threshold)	Self-consumption optimal when capital > TND 8M	Capital dynamically shapes component size and energy policy

technology selections, profitability, and environmental efficacy. By correlating the magnitude of initial investment with grid sale or self-consumption scenarios, we demonstrated that specific energy strategies, such as total self-consumption, become increasingly favorable beyond a particular investment threshold. This multi-criteria analysis offers a decision-making instrument tailored to the financial realities of sector players (individuals, developers, governmental entities), representing a novel contribution to the literature.

6. Conclusion

This study presents a detailed framework for optimizing grid-connected photovoltaic (PV) systems by incorporating investment constraints and evaluating various energy-selling scenarios. Unlike earlier research, this approach integrates the impact of initial capital on system sizing, enabling a more practical financial evaluation. By utilizing HOMER for system optimization and MATLAB for in-depth economic and environmental analysis, the study assesses multiple configurations to determine the most cost-effective and sustainable solution. The findings reveal that higher initial capital investment significantly lowers the Net Present Cost (NPC), Cost of Energy (COE), and CO₂ emissions. Among the evaluated scenarios, the self-consumption model (0 % energy sold to the grid) proved to be the most economically advantageous, achieving a COE of 0.130 TND/kWh and an NPC of TND 9.255 million. This configuration also resulted in a 41 % reduction in CO₂ emissions compared to conventional grid-based energy use. These outcomes

**Fig. 11.** Annual energy production of proposed system sources.

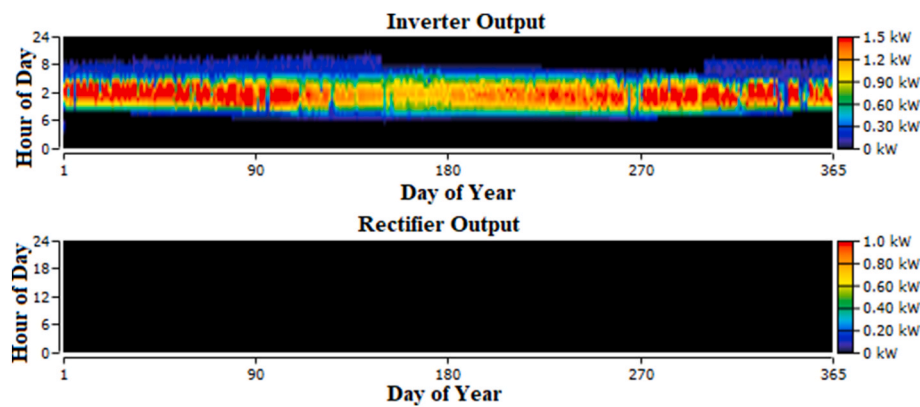


Fig. 12. Energy within one year when power supplied to grid (top) and when power consumed from grid (below).

underscore the financial and environmental benefits of prioritizing self-consumption over selling energy to the public grid.

Moreover, the study underscores the importance of strategic investment planning in PV system deployment. For large-scale projects, an investment exceeding TND 8 million ensures long-term profitability, reducing energy costs while maximizing renewable energy utilization. These insights are particularly relevant for policymakers, energy investors, and grid operators aiming to increase solar energy adoption in Tunisia and other regions with similar energy policies. Future research could explore advanced control strategies, such as AI-based energy management systems, to optimize real-time energy distribution and improve grid stability. Additionally, integrating demand-side management techniques and enhancing battery storage optimization could further boost the financial returns and operational efficiency of PV systems. By addressing the economic and environmental challenges of PV system integration, this study advances sustainable energy solutions and offers a valuable decision-making tool for stakeholders in the renewable energy sector.

CRedit authorship contribution statement

Gaith Baccouche: Writing – original draft, Validation, Software, Resources, Methodology, Investigation. **Sihem Amara:** Validation, Software, Methodology. **Chokri Ben Salah:** Validation, Supervision, Methodology. **Abdelaziz Salah Saidi:** Writing – review & editing. **Mehdi Tlija:** Writing – review & editing, Supervision, Methodology. **Abdelhamid Rabhi:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

This manuscript has not been published and is not under consideration for publication elsewhere. We have no conflicts of interest, but we do respectfully request that Pr. Maher Chaabene and DR. Abba Khiar-eddine not review our manuscript. If you feel that the manuscript is appropriate for your journal.

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Data availability

The data that has been used is confidential.

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